Retirement and the Demographic Transition: a Survey Experiment

Nello Esposito*
University of Naples Federico II and the Central Bank of Ireland

[Updated regulary. Click here for the latest version]

Abstract

Do individuals incorporate information about the national pension system into their retirement expectations and investment decisions? Using a novel representative sample of working Italians aged 15 to 75, this paper exploits exogenous variation in information by randomly exposing half of the sample to news about the pension system. I show that, on average, treated individuals revise their expected retirement age upward by 6 months and decrease their expected social security wealth by around 5%. Leveraging the Survey RCT as a first stage, I estimate the degree of substitutability between public pension wealth and private pension plans. I find very little substitutability, especially among individuals with low financial literacy.

JEL Classification: D12, D14, G51, E21, G51, G53, H55

Keywords: Retirement, Retirement Expectations, Demographic Decline, Survey Experiment

Acknowledgments: The author acknowledges funding by the European Union - Next Generation EU, part of the GRINS - Growing Resilient, Inclusive and Sustainable Project framework (GRINS PE00000018-CUPE63C22002140007). The views and opinions expressed are solely those of the author and do not necessarily reflect those of the European Union and are not the responsibility of the European Union. I am extremely grateful to my PhD supervisor, Tullio Jappelli, for invaluable assistance and guidance. I thank Luigi Pistaferri, Armando Miano, Chiara Lacava, Monica Langella, Annamaria Lusardi, Mario Padula, Marco Pagano, Marco Pagnozzi, Annalisa Scognamiglio, Lorenzo Pandolfi, Antonio Acconcia, Saverio Simonelli, Luca Picariello, Carla Guerriero, Pietro Maffettone, Gianluca Iaccarino, Antonia Pacelli, Carmine Russo, Nicol Barbieri, Alessandro Sciacchetano, Mattia Cenatiempo for useful comments and conversations. This paper reflects only my opinions and does not reflect the opinions of the Central Bank of Ireland.

*Email: nello.esposito@unina.it

"*No pension system is sustainable given today's demographic trends*"— Giancarlo Giorgetti, Italy's Finance Minister, July 18, 2024

1 Introduction

Italy's population is aging and shrinking. According to ISTAT, the National Bureau of Statistics, residents will drop by 5 million in 2050 and by 14 million in 2080 in comparison with 2023. The dependence ratio - computed as the number of over-65 citizens as a share of the 15- 64-year-old population - will rise to 0.73 from today's 0.57. At the same time, the economy has been sluggish in the last three decades: GDP grew by only 0.8% per year between 1990 and 2019, while the OECD average growth rate in the same period was 2.2%.

Slow economic growth and, most importantly, demographic decline pose a great threat to the sustainability of the first pillar of the pension system. As in most advanced countries, Italy's Social Security is based on the pay-as-you-go principle (Franco 2002): each year, benefits are financed by payroll taxes paid by those who are currently working. This intergenerational pact hinges on two factors: long-run economic growth and demographic trends (Diamond 1965). Therefore, given Italy's disappointing figures on both dimensions, the sustainability of the National pension system is a compelling issue.

Growing evidence shows that retirement expectations influence saving, investment, and labor supply decisions during working life (see Daminato and Padula 2024, Carta and De Philippis 2024, Jappelli et al. 2021). These retirement expectations, in turn, depend on how well workers know how the national social security system works and how aware they are of its financial conditions. Despite the importance of the subject, scant evidence shows how information about the pension system shapes these expectations. This gap in the literature likely stems from the difficulty of identifying plausibly exogenous information shocks to leverage for causal inference.

In this paper, using the novel Italian Survey of Consumer Expectations (ISCE), I provide an information treatment to a randomly formed half of the sample and examine how subjective retirement expectations respond. The information treatment consists of a paragraph sourced from the 2023 Annual Report of Italy's Social Security Agency about the financial difficulties of National Pension Systems in Europe. After the treatment, respondents answer two questions about the planned retirement age (the age at which they expect to retire) and the expected replacement rate (the ratio between the first public net pension income and the last net working stipend). Following Dominitz and Manski (2006), both questions are asked only to employed people in a probabilistic manner. Respondents are provided with real line intervals both for the age at which they plan to retire and for their expected replacement rate. They then assign a number between 0 and 100 to each interval, with the condition that all numbers add up to 100. For each individual in the survey, I obtain a subjective distribution for the planned retirement age and the expected replacement rate, from which I can compute the implied mean and standard deviation. Following Bottazzi et al. (2006), I compute a proxy for social security wealth (SSW),

defined as the discounted sum of all expected pension benefits each respondent will receive.

I find that providing information on the National Social Security system induces a sizable revision of retirement expectations. In particular, expected retirement age rises by approximately 6 months (0.45 years), the replacement rate decreases by almost 2 percentage points, and social security wealth drops by around €9000.

In the second part of the paper, I leverage the survey RCT as a first stage to investigate the substitutability between SSW private wealth and savings. I find little estimated substitutability between SSW and private pension plans. Considering all respondents in the sample, a reduction in SSW of €1000 makes them 0.42 percentage points more likely to hold a private pension plan. On the other hand, if the analysis is restricted only to employees - and therefore excluding the self-employed, who in principle might benefit from additional category-sponsored pension schemes - the degree of substitutability is 0.6, meaning that employees are 0.6 percentage points more likely to hold a private pension plan if SSW drops by €1000.

These results on low substitutability may stem from several factors. First, the heavy tax burden on wages in Italy (33% of payroll tax rate on gross earnings, of which 9% is paid by the employee and the remainder by the employer) prevents people from contributing sizable amounts to a private pension fund. Second, transaction costs - monetary and non-monetary - might be an additional reason, as recently argued by Chuukhmane and De Silva (2026). Another potential reason is financial literacy (Jappelli and Padula 2013 and Lusardi et al. 2017). Italy is among the advanced countries with the lowest level of financial literacy. Therefore, people may not buy these instruments because they do not understand how they work and don't trust them.

Related Literature. Social security shapes individuals' working, saving, and portfolio allocation decisions. Jappelli et al. (2021) document a positive correlation between subjective uncertainty about one's own pension and the holdings of private pension plans. Recent work by Carta and De Philippis (2024), Bianchi et al. (2023), and Brunello et al. (2023) looks at the impact of an increase in statutory retirement age on labor supply and career dynamics in Italy. Leveraging the 2011 pension reform as a quasi-experiment, they find that rising statutory retirement age induced women to work more - both on the intensive and the extensive margin - and hindered job promotion opportunities for younger workers, especially in the slowest-growing businesses. Attanasio and Brugiavini (2003) and Bottazzi et al. (2006), leveraging pension reforms implemented in Italy in the '90s, show that reducing the generosity of the system - both in terms of retirement age and entitlements – increases private savings and accumulated wealth. However, this replacement doesn't happen "one-to-one". Every euro of lost public pension wealth is replaced by less than one euro in private wealth, in contrast with the Permanent Income Hypothesis (Friedman 1957). Daminato and Padula (2024) replicate the empirical exercise by Bottazzi et al. (2006) and estimate a life cycle model for Italian retirees by matching the empirical estimates using indirect inference methods.

Despite the importance of the subject, little is known about how subjective retirement expec-

tations are influenced by actual knowledge of the PAYGO system and, in turn, how they influence retirement wealth. This question has not been answered in the literature because of the classical econometric concern of endogeneity. Indeed, retirement (and expectations about it) is linked to individuals' working histories, which are largely endogenous because of career choices (Guiso et al. 2002, Low et al. 2010 and Jappelli and Pistaferri 2017). Career choices, in turn, are linked to unobservable characteristics likely correlated with people's perceptions of their post-work life. Heterogeneity in risk aversion and in patience is the leading example of an omitted variable in the consumption/household finance literature (see Fuchs-Schündeln and Schündeln 2005 and Christelis et al. 2024 more recently). People with a lower distaste for working might be willing to work more years given the same expected wage profile. More patient people might decide to work more to "earn" a higher pension benefit and afford higher consumption in the retirement period. On the other hand, risk aversion impacts investment decisions and portfolio allocations (Merton 1969, 1971, Cocco et al. 2005), which in turn impact how people believe they will provide for their old age. Here, the randomized treatment induces exogenous variation in retirement expectations that is therefore uncorrelated with other characteristics influencing old-age beliefs.

Another limitation of the existing literature is the scant information on retirement expectations. Indeed, few existing datasets provide detailed individual and subjective distributions of planned retirement age and expected retirement rate. In the Italian context, Guiso et al. (2013) uses a representative survey of Italian investors to elicit pension uncertainty. In the survey, they ask three questions. First, they ask for the expected replacement rate, defined the same way as in this paper. Second, what is the minimum value, and third, what is the maximum value. Assuming either a triangular or uniform subjective distribution, they compute the implied standard deviation of the replacement rate for each individual and find that it correlates with age, job type, and sector. Jappelli et al. (2021) follows a similar approach. They use the 2016 SHIW wave, which asks the same type of questions as in Guiso et al. (2013), and find that pension uncertainty - measured by the subjective standard deviation of the replacement rate - correlates positively with the probability of having a private pension plan. Both papers rely on a nontrivial distributional assumption that may introduce measurement error in retirement expectations and uncertainty: either a triangular or a uniform distribution. The novelty of this paper lies in asking about retirement probabilistically, following the approach proposed by Manski (2004) and Dominitz and Manski (2006). Because each individual assigns a probability measure to a small interval on the line of natural numbers, we obtain a more accurate subjective distribution that requires no a priori distributional assumptions outside each bin. Indeed, the mean and standard deviation of the distribution, as shown in Appendix 1A, can be computed using the formulas for the expected value, the variance, and the standard deviation, only assuming that, within each bin, the probability mass is concentrated in the midpoint.

The closest paper is Mastrobuoni (2011), which examines the effect on retirement expectations of information contained in retirement statements sent to prospective retirees by the SSA in the United States in 1995. He finds that, although the treatment is informative and people

correctly receive the information in the letters, they do not change their expected retirement age or their decisions to claim Social Security benefits. My paper differs in two ways, particularly in its identification strategy. First, the result is the opposite: I find that people actually change their expected retirement age. Second, I also examine the treatment's impact on financial investment decisions, while Mastrobuoni (2011) looks only at whether individuals who receive the SSA statement claim Social Security benefits.

Survey RCTs have gained popularity in recent years because they can induce an exogenous information shock that can be leveraged for causal inference. Costa-Ramón et al. (2026) show that part-time female teachers in Switzerland severely downplay the long-term consequences of reduced labor supply. Therefore, randomly assigning them to financial information about their pension increases their working hours by 7%. Christelis et al. (2024) use a randomized assignment to hypothetical lottery wins, showing that wealth shocks increase stock holdings, but only at very high lottery-win levels. Kumar et al. (2023) and Coibion et al. (2024) investigate the effects of GDP growth uncertainty on firms' investment and households' expenditure, respectively. The main finding is that macro uncertainty induces firms to invest less (especially in R&D) and households to engage more in precautionary saving. Coibion et al. (2023) study the effect of news about inflation expectations on consumption decisions using a representative survey of Dutch households, documenting a positive effect on durable spending following the exogenously generated variation in inflation expectations. Coibion et al. (2023) look at the impact of the Fed's forward guidance on consumers' inflation and mortgage rate expectations. Georgarakos and Kenny (2022) document that an increase in people's perception of the adequacy of the policies implemented during the COVID-19 pandemic by the European Authorities spurred durable consumption. Finally, Coibion et al. (2021) find that news about the future Government's' debt causes households to anticipate a large increase in inflation, both on a 12-month and a 10-year horizon. All the papers cited above elicit firms' and consumers' subjective expectations probabilistically: asking firms to assign a probability to some real line intervals (Manski 2004 and Dominitz and Manski 2006). This paper is related to the last paper in both methodology and topic, even though it addresses a narrower type of fiscal sustainability problem: social security.

The last strand of the literature I connect to examines households' low participation in the stock market. A recent paper by Christelis et al. (2026) documents lower stock market participation by European households than in the United States. They also show that the main reasons are the lack of resources to invest, the risk, and low trust in the financial market. Other recent papers have pointed to regulatory frictions (Parker et al. 2025), transaction and mental costs (Chuukhmane and De Silva 2026) and income risk counter cyclicality (Catherine 2021). Other papers point to the importance of financial literacy (Jappelli and Padula 2013 and Lusardi et al. 2017). I add to this literature by showing that even though people know their public pension will be extremely low, they remain reluctant to invest in private pension plans, and that financial literacy varies across individuals. Therefore, financial literacy is a powerful tool that could help address not only inequality but also pension poverty.

Outline of the Paper. The Paper continues as follows. Section 2 briefly describes Italy's pension system, highlighting its sustainability issues. Section 3 describes the ISCE survey and the experimental design. Section 4 discusses the effects of the information treatment on retirement expectations. Section 5 estimates the substitutability between SSW and private pension plans. Section 6 examines the importance of financial literacy for the results, while Section 7 concludes and draws policy implications.

2 The Italian Pension System

This section provides the reader with key figures on the Italian pension system and its history since the 1970s. The Italian pension system has historically been extremely generous (Boeri and Rizzo 2020). Starting from the mid-'60s, reforms were implemented that reduced the eligibility criteria and facilitated access to retirement[1]. As a result, pension spending soared in subsequent decades, accounting for a larger share of national output. Indeed, social security spending rose from 5.3% of GDP in 1960 to 15.5% in 1986 (Galli and Masera, 1988). In fact, this astonishing increase in retirement-related spending is considered one of the main drivers of Italy's public debt rise in the 70s and the '80s (still Galli and Masera 1988 and Rossi 2020).

During the '90s, many reforms were implemented to curb this upward trend. In 1992, the Amato[2] reform lowered pension benefits and raised the eligible retirement age for workers who had accumulated, as of 1993, less than 15 years of contributions. This intervention was followed by a more stringent reform in 1995 that modified the algorithm used to compute pension benefits. The transition shifted from an earnings-based formula, in which the benefit is proportional to the average income earned during the last years of work, to a contribution-based formula, in which benefits are proportional to all payroll taxes paid over the entire working period. Because wages typically increase with age and experience, this intervention entailed a massive reduction in benefits for younger cohorts of workers (Bottazzi et al. 2006).

The most important reform in recent years was implemented in 2011. The Fornero reform, named after the Labor Minister Elsa Fornero, raised the retirement age to 67 and anchored it to average life expectancy. Alternatively, women and men can retire, regardless of age, once they have accumulated at least 41 or 42 years of contributions, respectively.

In 2018, *Quota 100* was introduced. Under Q100, people could retire at least at 62 years of age with 38 years of contributions[3], a significantly more generous retirement regime than the Fornero one. Q100 was phased out in 2021, replaced by a slightly more stringent regime called Q102 (meaning $62 + 38$) for 2022 and 2023, then abolished definitively in 2025.

[1] A famous reform is the 1973 Rumor reform, named after Prime Minister Mariano Rumor. The reform stated that public sector workers could retire as soon as they reached 15 years of contribution. This gave rise to the "baby retirees", i.e., people who stopped working at the age of 50, 40, or even younger. Anecdotal evidence can be found in Boeri and Rizzo (2020).

[2] Named after Prime Minister Giuliano Amato

[3] 62 years of age + 38 years of work; that is where Q100 comes from.

Today, the pension system is one of the hottest topics in Italian politics, making headlines every day, as the quote at the beginning of the paper shows. And it's rightfully so, given the severe demographic crisis Italy faces, as the introduction's ISTAT-based data show. Pension spending accounts for roughly 15% of GDP. It is expected to peak at 17% in 2045 (INPS 2024), making Italy's Social Security the most generous among OECD countries, as OECD (2023) documents. Still, according to INPS (2024), the General Government transferred more than €160 bn to INPS, two-thirds of the revenues collected by the personal income tax (around €235 bn). Although part of these transfers is meant to fund social assistance rather than pensions, it remains an impressive figure that speaks to the system's sustainability issues.

3 The Survey and Experimental Design

In this section, I describe how the survey and the information treatment are conducted. I also present some descriptive evidence. In the next two sections, instead, I discuss the treatment effects. The experiment uses the novel Italian Survey of Consumer Expectations (ISCE). Started in October 2023 (first wave), ISCE is a quarterly survey that interviews a rotating representative panel of 5000 Italians aged 15 to 75. The survey asks questions about consumption and income expectations, as well as retirement and expectations about macroeconomic variables such as GDP growth, inflation, unemployment, and interest rates. Guiso and Jappelli (2024) provide a detailed description of the survey. Appendix A shows the entire surevy.

The Survey. In the first part, the survey asks for basic demographic information on age, gender, place of residence, education, and employment. It then asks income- and work-related questions, as well as questions on wealth. It asks about family and individual net monthly income. It also includes questions on perceived employment risk (for the employed) and job-finding probability (for the unemployed or those looking for their first job). It asks about homeownership and net wealth, computed as real assets plus financial assets minus debts. It also asks about the composition of the financial portfolio, but only on the extensive margin, i.e., whether people own a certain financial product. The wording of the questions is as follows:

"Think about your financial investments. Which of the following savings/investment products do you own? (Yes or No for each financial instrument)"

- *A Bank Account*
- *Fixed Income Securities (Government Bonds or Private Bonds)*
- *Stocks and Equity Funds*
- *A supplementary pension plan*

- *Life Insurance*

At this point in the third wave (April 2024), the sample is randomly split into two groups that I call C and T. Before continuing the survey, group T reads the following sentence sourced from the 2023 Annual INPS Report:

"According to the latest Annual Report of INPS, the progressive aging of the population, which increases the number of retirees and reduces the number of active workers, makes it increasingly difficult to finance pensions in all European countries, raising financial stability issues."

This treatment has a twofold aim. First, it aims to show that the Italian PAYGO has struggled in recent decades because of population aging. In the final part, the treatment also notes that many European countries may face financial stability issues as social security becomes harder to finance. As discussed by Jappelli et al. (2021), pension risk comprises three components. The first is idiosyncratic income risk: people have different career paths and working histories, some shaped by choices and others by unemployment or health shocks. Second, as discussed in the first introductory part, there are long-run risks from GDP growth and demographic projections. Third, changes in regulation could affect both retirement age and retirement income. This treatment speaks to the last two sources of risk: the reference to "financial stability issues" points to both. Group C doesn't read any sentence, but continues the survey without any additional information.

After the treatment, both groups answer two questions about retirement. The former is on planned retirement age:

"At what age do you plan to retire?"

- *Before 58 years of age*
- *Between 58 and 60*
- *Between 60 and 62*
- *Between 62 and 64*
- *Between 64 and 66*
- *Between 66 and 68*
- *Between 68 and 70*
- *Over 72*

Total*.....*100

The latter is on the public replacement rate, defined in the question as the ratio between the public pension check and labor earnings. Respondents are asked to exclude retirement income from sources other than the Government, to address the fact that many people in the sample have private pension plans they will earn once retired. The phrasing of the question is:

"Think about when you will retire and consider only the public pension, i.e., exclude any pension funds and supplementary pensions. What percentage of your earned income will the government pension represent? "

- *Less than 40% of your last salary before retirement*
- *Between 40 and 50% of the last salary received before retirement*
- *Between 50 and 60% of the last salary received before retirement*
- *Between 60 and 70% of the last salary received before retirement*
- *Between 70 and 80% of the last salary received before retirement*
- *Between 80 and 90% of the last salary received before retirement*
- *Between 90 and 100% of the last salary received before retirement*

Total*.....*100

This type of question lets us elicit a subjective distribution for the variable in question, i.e., a distribution for each individual over the proposed intervals. We can compute the moments of these subjective distributions easily. Indeed, assuming that, within each bin, the probability mass is concentrated at the midpoint, the mean of the distribution is simply the weighted average of the midpoints, using the scores assigned by the respondent to each range as weights. The variance and standard deviation are then computed using the well-known formulas shown in Appendix B.

Descriptive Evidence. Table 1 presents summary statistics and compares the wave 2 cross-section of the ISCE with the latest available 2022 Bank of Italy SHIW. Both surveys include employed people aged between 15 and 75. Both surveys use sample weights. Overall, ISCE does a good job of representing the population. The average age is about 44 years, 58 percent of respondents are male, and about one-half are married. One third of the sample lives in the South (and on the Islands), while one half is surveyed in the North. Education levels look fairly in line with the 2022 SHIW: 27 percent of the sample holds a Master's Degree or more, exactly as in

SHIW. Interestingly enough, in ISCE, people report an expected replacement rate of 61 percent (65 in SHIW), while the planned retirement age is similar in both surveys: 66 years old.

Table 2, on the other hand, shows the summary statistics of the sample I use for the causal analysis. I focus on employed individuals (employees and self-employed, including public sector workers) aged 50 or younger and on individuals I observe both in wave 2 and wave 3. As in every RCT, it's important to ensure the sample was randomly split and that the treatment and control groups are comparable. To this purpose, I compare key observable characteristics in the two groups in the wave preceding the treatment, i.e., wave 2. Overall, randomization worked well, as most differences in observables between the two groups are neither economically nor statistically distinguishable from 0. The only exception is geographical residence: while the share of residents of the North is the same, the share of the South is 5 percentage points lower in the treatment group (and the share of the Center is 4 percentage points higher). This might happen when the randomization is done, as in this case, on a small number of individuals.

For expositional convenience, I will refer to the implied mean of the subjective distribution of planned retirement age as planned retirement age. By the same token, I will refer to the implied standard deviation of the subjective distribution of planned retirement age as the standard deviation of retirement age. I will apply this terminology to the replacement rate as well. So, the implied mean of the subjective distribution of expected replacement rate will be referred to as expected replacement rate. In contrast, I will refer to the implied standard deviation as the standard deviation of the replacement rate. Following this terminology, Figure 1 shows the cross-sectional distribution of the expected retirement age (top panel) and the expected replacement rate (bottom panel). For the former, the most striking finding is that a share of respondents (more than 10%) expect to retire above age 70, the maximum working age in Italy. In other words, many respondents expect to work longer than today's maximum age, perhaps because they expect it to be extended by the time they become eligible for retirement. Looking at the expected replacement rate (bottom panel), 10% of respondents expect to receive as much as 35% of their last working income from the public pension fund. In contrast, almost no one expects full replacement (95% of the last working income). Figure 2 shows the same graphs for the standard deviation of retirement age and replacement rate. A significant share of respondents (one third) are certain about their retirement outcomes, as they assign 100% probability to a single bin.

Figure 3 shows the correlation between the first and second moments. The left panel shows a strong negative correlation between expected retirement age and replacement rate. The right panel, by contrast, shows that uncertainty about retirement age and replacement rate is strongly and positively correlated. This means that if a person is very uncertain about when she will retire, she is also very likely to be uncertain about how much she will get from the Social Security fund.

Expected Social Security Wealth Given the information on subjective expectations on retirement age and pension benefits, and following the approach by Bottazzi et al. (2006), I can

compute a proxy of expected Social Security wealth, i.e., the discounted value of all the Social Security benefits people expect to receive from their retirement to their death. The formula I use to compute such a measure of Social Security wealth (SSW) is the following:

$$SSWY_t = \left[p(N_t|t) \left(\frac{1+g_y}{1+r} \right)^{N_t - t} R_t \right] \left[\sum_{\tau=N_t}^{T} p(\tau|N_t) \left(\frac{1+g_R}{1+r} \right)^{\tau - N_t} \right] \tag{1}$$

In equation (1), N_t is the expected retirement age, R_t is the expected replacement rate, and T is the terminal age, assumed to be equal to 119[4]. The intuition of equation (1) is the same as for any financial asset: the value of social security equals the expected cash flows it generates, discounted to the current period. The second square bracket is the sum of all the replacement rates expected by people - assumed to grow at a constant rate $g_R = 0$ - from the year of retirement (N_t) until death, adjusted by the conditional survival probability of living from N_t until each year in the summation. The first square bracket, by contrast, is the discount factor, adjusted by the probability of surviving from the current age (t) until the year of retirement. This computation assumes an annual interest rate of 3% and an average annual income growth rate of 0.5%, consistent with long-run projected productivity (and wage) growth in Italy[5]. Figure 4 shows the distribution of the expected SSW over income. I have computed using (1). The average SSW/Y is 7.3, with a standard deviation of 2.383. Figure 5 shows the correlation between this measure of social security wealth and private financial wealth. The correlation is negative (-0.231) and statistically significant at the 99% level. However, it's substantially lower (in magnitude) than -1, as predicted by the permanent income hypothesis (Friedman 1957).

4 Treatment Effects

Retirement Expectations. In this section, I look at the effect of the information treatment on the planned retirement age and on the expected replacement rate. To this end, I estimate the following equations:

$$\begin{aligned} Y_i &= \alpha + \beta Treat_i + \varphi \mathbf{X}_i + \varepsilon_i \\ Y_i &= \alpha + \beta_1 Treat_i + \gamma Prior_i + \sigma Treat_i \times Prior_i + \varphi \mathbf{X}_i + \varepsilon_i \end{aligned} \tag{2}$$

In equation (2), the dependent variables are the expected retirement age (N_i), the expected replacement rate (R_i), and the expected Social Security wealth-to-income SSW (its absolute value in €1000, its ratio to income and its log). Then, $Treat_i$ is a dummy equal to 1 if the individual received the information in the survey, 0 otherwise. $Prior_i$ is the same variable as Y_i (either N_i

[4]The Italian Statistical Agency assigns a positive probability of living up to 119 years. I have checked that changing the terminal age to more "realistic" values does not change the results.

[5]These computations are both qualitatively and quantitatively unaffected when considering different assumptions on wage growth.

or R_i), but elicited in the wave preceding the experiment. X_i is a vector of controls. Given that the treatment is randomized, it is, by construction, independent of any observable characteristic affecting the dependent variable. Nevertheless, I include controls to improve the precision of the estimate of β, the coefficient of interest, which measures the change in retirement expectations caused by the treatment. In the second equation, the coefficient γ measures the correlation between the posterior and the prior expectation for the controls. It should be positive and highly significant, as it's sensible that people's expectations are highly correlated over three months. The coefficient β_1 captures the "level" component of the causal effect. The coefficient σ, on the other hand, could be seen in two ways. First, from a Bayesian perspective, it is how much the treatment reduces the importance of the prior and should be negative. As new information comes in (the treatment), the respondent should pay less attention to his/her prior if the treatment is informative. The second interpretation of σ is how much the treatment effect varies as a function of the prior. Intuition suggests it should be negative for the retirement age: if, ex ante, I expected to retire late, the treatment should not change my expectation much. Finally, ε_i is a mean-zero error term. Because the treatment is randomized, the strict exogeneity assumption holds, and the estimated OLS coefficient is unbiased.

The estimation results of (2) are showed in Table 3 and Table 4. The first three columns of Table 3 show the effect on the expected retirement age, while the second three columns show the expected replacement rate. The treatment raises the expected retirement age by 0.5 years (6 months). As expected, adding a control does not affect the point estimate and only slightly improves precision, as the standard error decreases from 0.21 to 0.20. Column 3 shows the estimated coefficients of an equation in which the treatment dummy is interacted with the prior belief on retirement age. The coefficient on the interaction term is negative, suggesting that the treatment effect decreases with pre-treatment expectations. From a Bayesian perspective, this means the treatment is somewhat successful at conveying new information, as treated individuals place less weight on their prior beliefs. From a purely econometric perspective, however, it means the treatment effect decreases in the prior. Figure 6 plots the total effect as a function of the prior belief. The figure delivers two insights. First, people who, before treatment, already expected to retire very late (say, in their 70s) react to the treatment by anticipating their expected retirement age. I attribute this result to the following reasons. Expecting to retire very late might signal a preference for work. These people presumably like their job and are happy to postpone retirement. However, the treatment "scares" them and induces them to anticipate their retirement plans. Indeed, the treatment points to possible financial stability issues in the pension system, increasing the perceived risk of a pension reform that might reduce their pension benefits. In other words, even though some respondents are happy to continue their job and retire later in age, they start considering the political risk (Jappelli et al. 2021) of pension reforms that might occur in the last years of their career[6]. The second insight is that the total effect on expected

[6]It's worth noting that something like this happened in 2012 after the Fornero reform. Many people who were only a few years away from retirement decided to quit their jobs and wait for eligibility while living off

retirement age is close to 0 at 67 years old, that is, the minimum retirement age as of 2026. The second half of Table 1.3 shows the results for the replacement rate. The treatment induces a downward revision of the expected replacement rate of almost 2 percentage points, and nothing changes when you add controls. There is, however, no heterogeneity in the effect across prior beliefs about the replacement rate.

Expected SSW Table 4 shows the treatment effect on the expected social security wealth measure computed as in equation (1). The first two columns (with and without controls, respectively) show the effect on the ratio of SSW to labor income. The effect is negative: treatment exposure reduces the SSW-to-income ratio by 0.50. In monetary terms, this amount corresponds to €9000, i.e., an overall effect of 5% compared to the control group (columns 3 and 4).

5 Substitutability between SSW and Private Wealth

In the previous section, I examined how providing information on the sustainability of the pension system affects retirement expectations. In this section, I leverage the survey RCT to estimate the degree of substitutability between social security wealth and several forms of private wealth, especially private pension plans. In particular, I estimate the following equation using instrumental variables:

$$Holds\ Private\ Pension_{it} = \alpha + \rho Z_{it} + \varphi \mathbf{X}_{it} + \lambda_t + v_{it} \tag{3}$$

In equation (3), $Holds\ Private\ Pension_{it}$ is a dummy equal to 1 (multiplied by 100) if i holds a private pension plan at time t, while λ_t are monthly FEs. The first stage is:

$$Z_{it} = \alpha + \beta Treat_i + \varphi \mathbf{X}_i + \lambda_t + u_i \tag{4}$$

In equations (3) and (4), $Z_{it} \in \{\mathbb{E}[SSW_i], \mathbb{E}[SSW/Y_{it}]\}$ the term λ_t represents wave (quarters) FEs, while X is the usual vector of controls. Given that, in wave 3, information on the composition of the financial portfolio is collected *before* the treatment, I estimate these equations using only the after-treatment waves, i.e., starting from wave 4. The coefficient of interest is ρ, capturing the causal effect of a reduction in expected SSW on the propensity to own a private pension plan among compliers, defined in this context as those who changed their SSW expectations because of exposure to the information treatment. The coefficient of interest is ρ, which captures the LATE of a decrease in expected social security wealth for the compliers, who in this case are defined as those who revised their retirement expectations *because* of the information they received.

While Table 5 shows the first stage, Table 6 shows the estimated coefficients of (3), using as Z_{it} the level of SSW. Column 1 reports the effect in the full sample and says that if SSW falls

unemployment insurance. However, the Government raised the retirement age for them by several years higher than the total duration of the unemployment insurance, leaving them unemployed at a very advanced age.

by €1000, survey respondents are 0.43 percentage points more likely to hold a private pension plan. Compared with the baseline average participation of 30%, this effect corresponds to a 1.4% increase in participation. Column 2, by contrast, reports the effect estimated only on employees, excluding the self-employed. The self-employed typically work longer than employees, and sometimes have access to category-based pension funds that are not accessible to employees. In addition, below a certain income threshold, independent workers have access to much more favorable fiscal and contribution treatment, which could encourage them to invest more in private pensions. That's why it's interesting to focus only on employees. Column 2 suggests that if SSW goes down by €1000, employees are 0.60 points more likely to hold a private pension plan, corresponding to almost 2% of the baseline mean. Columns 3 and 4 show results using the ratio, rather than the absolute value, as the endogenous variable. The message is the same: the effect on all workers is small and borderline significantly different from 0, while the effect on employees is much stronger, both economically and statistically. In particular, the fourth column implies that when the SSW-to-Y ratio increases by 1, households are 12 percentage points more likely to hold a pension plan.

The main takeaway of the empirical analysis is that workers, even though they respond to the information treatment by revising their expected social security wealth, do not meaningfully increase the demand for a private pension plan. Indeed, as anticipated by Figure 5, their substitution between social security wealth and private wealth is far from being perfect, as predicted by the standard permanent income hypothesis[7]. The literature has proposed several explanations for this puzzle. The first explanation concerns transaction costs. A recent work by Chuukhmane and De Silva (2026) shows that transaction costs explain better than preferences - i.e., risk aversion - the low stock market exposure in 401(k) plans in the United States. The second potential explanation is given by the fact that pension plans are highly illiquid assets, and so, while they are very good instruments for serving the intertemporal motive of saving, they are not well-suited to absorb income fluctuations (Auclert et al. 2024, Graves 2025). In other words, a trade-off may exist between the two main motives for saving, and people may prefer liquid assets to insure against short-term income fluctuations. This becomes even more relevant if one assumes that income shocks are cyclical with respect to the stock market. Given that a sizable share of pension funds is invested in the stock market, a stock market downturn might be more likely to be associated with income shocks (Catherine 2021). Finally, it is the case that very few households are familiar with basic financial instruments such as bonds and stocks, and this might keep them from investing - directly or indirectly - in the stock market. Lusardi et al. (2017) show that heterogeneity in financial literacy explains a lot of the cross-sectional wealth inequality observed in the United States.

While this empirical analysis cannot investigate the role of transaction costs and income risk, ISCE contains information on interviewees' financial literacy, allowing me to examine how the

[7]This result is not new to the literature: Attanasio and Brugiavini (2003) and Bottazzi et al. (2006) have estimated for Italy very little substitutability

effect varies across levels of financial literacy. This is the path I pursue in the following section, where I investigate heterogeneity in the estimated effect across interesting subgroups of the sample.

6 The Role of Financial Literacy

The average results explained above vary substantially across levels of financial literacy. Previous papers (see Jappelli and Padula 2013 and Lusardi and Mitchell 2011) have shown that financial literacy plays a key role in investment decisions. In fact, financial literacy helps explain cross-sectional wealth inequality in the United States (Lusardi et al. 2017). The intuition is that higher financial literacy allows households to access better investment opportunities. This changes the Euler Equation people face, as the interest rate at which they can invest their money is higher.

In the survey, we measure financial literacy as in Lusardi (2008) using the "Big Three" questions. The questions evaluate people's understanding of the difference between nominal and real variables, the compound interest rate, and portfolio diversification. The questions are as follows.

"Imagine leaving $100 in a bank account that yields an interest rate of 2 % per year and has no fees. After 5 years, how much do you imagine is available?"

- ***More than 102 euros***
- *Exactly 102 euros*
- *Less than 102 euros*
- *I don't know*
- *I'd rather not answer*

"Suppose you leave 1,000 euros in a checking account that yields an interest rate of 1 percent and has no management fees. Imagine, too, that inflation is 2 percent. Do you think that, in a year, when you withdraw the money, you will be able to buy the same amount of goods that you could buy by spending the 1,000 euros today?"

- *Yes*
- ***No, I will be able to buy a smaller quantity***
- *No, I will be able to buy a larger quantity*
- *I don't know*

- *I'd rather not answer*

"In your opinion, does buying shares in a single company usually provide a more secure return than buying shares in multiple companies through a mutual fund?"

- *True*
- ***False***
- *I don't know*
- *I'd rather not answer*

I consider *"I don't know"* and *"I'd rather not answer"* as wrong answers[8], and I group respondents in the low or high literacy group if they answered correctly to only one question (or zero) and above one, respectively. Table 7 shows the estimation results for a reduced-form equation, with the sample split by financial literacy level. The odd columns show results for the subsample with low financial literacy, while the even columns show results for the "high" financial literacy group. The point estimates are higher for the second group: only high-financial-literacy respondents increase their ownership of private pension plans if they receive the information treatment, especially among employees.

7 Conclusions

Using the novel Italian Survey of Consumer Expectations, I perform a survey RCT in which I provide half of the sample with information on the troubled financial status of the National Social Security System. Treated households revise their retirement expectations by about 6 months and reduce their expected Social Security wealth by about 5%. Leveraging the information treatment as a first stage, I estimate the degree of substitutability between public pension wealth and private wealth. I find a very small degree of substitutability between the two asset types. The effect is somewhat stronger for employees, while it's completely absent for the self-employed. Finally, I show that people with high financial literacy mostly drive the increase in private pension plans, measured using the *big three* questions as in Lusardi (2008).

This paper carries two interconnected policy-relevant messages. First, more work is needed to increase public awareness that population aging is putting pressure on public finances. The underlying intuition is that if people knew about the pension system's difficult financial situation, they would not have revised their retirement expectations significantly, since they would have already factored in that information. The second message is on the importance of financial

[8]The reason is that these three questions are very simple for those who are a little bit familiar with these concepts. If, on the one hand, a person who answers well to all the questions cannot by any means be deemed an expert, those who fail to answer at least one correctly are not experts for sure

literacy. While the literature has identified financial literacy as a key driver of cross-sectional wealth inequality, it may also be an important tool for addressing population aging and low pensions. If individuals become aware that their pension will be a very low percentage of their working income, they need to start saving for retirement from the beginning of their working life. To do that, they need the right financial knowledge.

References

Attanasio, O. P. and A. Brugiavini (2003). Social security and households' saving. *the Quarterly Journal of economics 118*(3), 1075–1119.

Auclert, A., M. Rognlie, and L. Straub (2024). The intertemporal keynesian cross. *Journal of Political Economy 132*(12), 4068–4121.

Bianchi, N., G. Bovini, J. Li, M. Paradisi, and M. Powell (2023). Career spillovers in internal labour markets. *The Review of Economic Studies 90*(4), 1800–1831.

Boeri, T. and S. Rizzo (2020). *Riprendiamoci lo Stato: Come l'Italia può ripartire*. Feltrinelli Editore.

Bottazzi, R., T. Jappelli, and M. Padula (2006). Retirement expectations, pension reforms, and their impact on private wealth accumulation. *Journal of Public Economics 90*(12), 2187–2212.

Brunello, G., M. De Paola, and L. Rocco (2023). Pension reforms, longer working horizons and absence from work. *Longer Working Horizons and Absence from Work*.

Carta, F. and M. De Philippis (2024). The forward-looking effect of increasing the full retirement age. *The Economic Journal 134*(657), 165–192.

Catherine, S. (2021, 12). Countercyclical labor income risk and portfolio choices over the life cycle. *The Review of Financial Studies 35*(9), 4016–4054.

Christelis, D., D. Georgarakos, T. Jappelli, and G. Kenny (2024). Wealth shocks and portfolio choice. *Journal of Monetary Economics*, 103632.

Christelis, D., D. Georgarakos, T. Jappelli, G. Kenny, and J. Meyer (2026). Stockholding in europe: Evidence from the consumer expectations survey.

Chuukhmane, T. and T. De Silva (2026). What drives investors' portfolio choices? separating risk preferences from frictions. *The Journal of Finance 81*(1), 5–48.

Cocco, J. F., F. J. Gomes, and P. J. Maenhout (2005). Consumption and portfolio choice over the life cycle. *The Review of Financial Studies 18*(2), 491–533.

Coibion, O., D. Georgarakos, Y. Gorodnichenko, G. Kenny, and M. Weber (2024). The effect of macroeconomic uncertainty on household spending. *American Economic Review 114*(3), 645–677.

Coibion, O., D. Georgarakos, Y. Gorodnichenko, and M. van Rooij (2023). How does consumption respond to news about inflation? field evidence from a randomized control trial. *American Economic Journal: Macroeconomics 15*(3), 109–52.

Coibion, O., D. Georgarakos, Y. Gorodnichenko, and M. Weber (2023). Forward guidance and household expectations. *Journal of the European Economic Association*, jvad003.

Coibion, O., Y. Gorodnichenko, and M. Weber (2021). Fiscal policy and households' inflation expectations: Evidence from a randomized control trial. Technical report, National Bureau of Economic Research.

Costa-Ramón, A., M. Slotwinski, U. Schaede, and A. A. Brenøe (2026, 05). (not) thinking about the future: Financial information and maternal labor supply. *The Quarterly Journal of Economics 141*(2), 1335–1382.

Daminato, C. and M. Padula (2024). The life-cycle effects of pension reforms: a structural approach. *Journal of the European Economic Association 22*(1), 355–392.

Diamond, P. A. (1965). National debt in a neoclassical growth model. *The American Economic Review 55*(5), 1126–1150.

Dominitz, J. and C. F. Manski (2006). Measuring pension-benefit expectations probabilistically. *LABOUR 20*(2), 201–236.

Franco, D. (2002). Italy: a never-ending pension reform. In *Social security pension reform in Europe*, pp. 211–262. University of Chicago Press.

Friedman, M. (1957). The permanent income hypothesis. In *A theory of the consumption function*, pp. 20–37. Princeton University Press.

Fuchs-Schündeln, N. and M. Schündeln (2005, 08). Precautionary Savings and Self-Selection: Evidence from the German Reunification "Experiment"*. *The Quarterly Journal of Economics 120*(3), 1085–1120.

Galli, G. and R. S. Masera (1988). Government deficits and debts. the necessity and cost of adjustment: the case of italy. In *Keynes and Economic Policy: The Relevance of the General Theory after Fifty Years*, pp. 175–215. Springer.

Georgarakos, D. and G. Kenny (2022). Household spending and fiscal support during the covid-19 pandemic: Insights from a new consumer survey. *Journal of Monetary Economics 129*, S1–S14.

Graves, S. (2025, April). Does unemployment risk affect business cycle dynamics? *American Economic Journal: Macroeconomics 17*(2), 65–100.

Guiso, L. and T. Jappelli (2024). The italian survey of consumer expectations: Statistical bulletin. Technical report, Centre for Studies in Economics and Finance (CSEF), University of Naples, Italy.

Guiso, L., T. Jappelli, and M. Padula (2013). Pension wealth uncertainty. *Journal of Risk Insurance 80*(4), 1057–1085.

Guiso, L., T. Jappelli, and L. Pistaferri (2002). An empirical analysis of earnings and employment risk. *Journal of Business & Economic Statistics 20*(2), 241–253.

INPS (2024). Xxiii rapporto annuale.

Jappelli, T., I. Marino, and M. Padula (2021). Social security uncertainty and demand for retirement saving. *Review of Income and Wealth 67*(4), 810–834.

Jappelli, T. and M. Padula (2013). Investment in financial literacy and saving decisions. *Journal of Banking Finance 37*(8), 2779–2792.

Jappelli, T. and L. Pistaferri (2017, 11). *The Economics of Consumption: Theory and Evidence*. Oxford University Press.

Kumar, S., Y. Gorodnichenko, and O. Coibion (2023). The effect of macroeconomic uncertainty on firm decisions. *Econometrica 91*(4), 1297–1332.

Low, H., C. Meghir, and L. Pistaferri (2010). Wage risk and employment risk over the life cycle. *American Economic Review 100*(4), 1432–1467.

Lusardi, A. (2008). Financial literacy: an essential tool for informed consumer choice? Technical report, National Bureau of Economic Research.

Lusardi, A., P.-C. Michaud, and O. S. Mitchell (2017). Optimal financial knowledge and wealth inequality. *Journal of political Economy 125*(2), 431–477.

Lusardi, A. and O. S. Mitchell (2011). Financial literacy and retirement planning in the united states. *Journal of pension economics & finance 10*(4), 509–525.

Manski, C. F. (2004). Measuring expectations. *Econometrica 72*(5), 1329–1376.

Mastrobuoni, G. (2011). The role of information for retirement behavior: Evidence based on the stepwise introduction of the social security statement. *Journal of Public Economics 95*(7-8), 913–925.

Merton, R. C. (1969). Lifetime portfolio selection under uncertainty: The continuous-time case. *The review of Economics and Statistics*, 247–257.

Merton, R. C. (1971). Optimum consumption and portfolio rules in a continuous-time model. *Journal of Economic Theory 3*(4), 373–413.

OECD (2023). Pensions at a glance 2023: Oecd and g20 indicators, oecd publishing, paris.

Parker, J. A., A. Schoar, A. Cole, and D. Simester (2025). Household portfolios and retirement saving over the life cycle. *The Journal of Finance 80*(5), 2739–2787.

Rossi, S. (2020). *La politica economica italiana dal 1968 a oggi*. Gius. Laterza & Figli Spa.

Tables

Table 1: Summary Statistics and Comparison with Shiw

	ISCE Wave 2		**SHIW 2022**	
	Mean	Std. Dev	Mean	Std. Dev
Age	44.06	11.15	45.83	11.93
Male	0.57	0.50	0.57	0.50
Married	0.52	0.50	0.58	0.49
Family Size	2.87	1.14	3.04	1.24
South	0.32	0.47	0.28	0.45
Centre	0.21	0.40	0.22	0.41
North	0.47	0.50	0.50	0.50
Primary Education	0.33	0.47	0.25	0.43
Secondary Education	0.42	0.49	0.08	0.27
Tertiary Education	0.25	0.43	0.68	0.47
Employee	0.83	0.38	0.81	0.39
Public sector	0.19	0.39	0.22	0.41
Personal Monthly earnings (in '000€)	1.64	1.25	1.32	1.47
Home Owner	0.77	0.42	0.76	0.43
Pension Replacement Rate (expected)	61.13	16.36	66.70	17.53
Retirement Age (expected)	65.76	3.77	66.28	3.77
Observations	**2556**		**8397**	

Notes: This table compares the ISCE sample with the 2022 wave of the Bank of Italy's SHIW, the latest available. Sample weights are applied to both surveys.

Table 2: Balance Test

	Controls	**Treated**	**Difference**	Std. Error
Age	38.20	37.85	0.36	(0.40)
Male	0.54	0.57	-0.02	(0.03)
Married	0.44	0.46	-0.02	(0.03)
Family size	2.94	2.84	0.10*	(0.06)
South	0.34	0.39	-0.05***	(0.03)
Centre	0.20	0.16	0.04*	(0.02)
North	0.46	0.45	0.01	(0.03)
Primary Education	0.23	0.21	0.01	(0.02)
Secondary Education	0.41	0.41	-0.01	(0.03)
Tertiary Education	0.37	0.37	-0.01	(0.03)
Employee	0.84	0.86	-0.02	(0.02)
Public Sector	0.20	0.19	0.01	(0.02)
Unemployment risk	0.16	0.17	-0.01	(0.01)
Family net monthly income (in €1000)	2.38	2.24	0.13*	(0.08)
Home owner	0.76	0.76	0.00	(0.02)
Total Wealth (in €1000)	166.17	163.25	2.92	(9.97)
Expected Replacement Rate	60.81	59.83	0.98	(0.86)
Expected Retirement Age	65.94	66.17	-0.23	(0.20)
Total Consumption (in €1000)	1.39	1.39	-0.01	(0.07)
Expected Consumption growth rate	0.72	0.78	-0.06	(0.20)
Expected Earnings growth rate	-0.69	-0.49	-0.20	(0.18)
Holds a Private Pension	0.29	0.30	-0.01	(0.02)
Observations	**740**	**710**		

Notes: This table compares the treatment and control groups to show they are comparable in observable characteristics.

Table 3: The Effect of the Treatment on Retirement Expectations

	Exp. Retirement Age			Exp. Replacement Rate		
	(1)	(2)	(3)	(4)	(5)	(6)
Treat	0.50**	0.46**	10.98***	-1.89**	-1.83**	-2.30
	(0.21)	(0.20)	(3.30)	(0.84)	(0.84)	(3.33)
Treat x Prior			-0.16***			
			(0.05)			
Prior Retirement Age			0.67***			
			(0.03)			
Treat x Prior Rep. Rate						0.01
						(0.05)
Prior Rep. Rate						0.39***
						(0.04)
Observations	1450	1450	1450	1450	1450	1450
Controls	·	✓	✓	·	✓	✓

Notes: This table show the estimation results of equations (2). The dependent variable is either the expected retirement age or the expected replacement rate. Standard errors in parentheses and * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: The Effect of the Treatment on Expected Social Security Wealth in Wave 3

	SSW/Y	SSW/Y	SSW (€1000)	$\log(SSW)$
	(1)	(2)	(3)	(4)
Treat	-0.46***	-0.44***	-9.10**	-0.05**
	(0.16)	(0.15)	(4.15)	(0.02)
Observations	1450	1450	1450	1450
Controls	·	✓	✓	✓

Notes: This table show the estimation results of equations (2). The dependent variable is the expected social security wealth, its log, or its ratio with income. Standard errors in parentheses and * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: First Stage

	SSW (in €1000)			SSW/Y		
	(1)	(2)	(3)	(4)	(5)	(6)
Treat	-7.04***	-8.54***	-6.61***	-0.39***	-0.38***	-0.43***
	(2.01)	(2.19)	(2.25)	(0.08)	(0.09)	(0.10)
Observations	4960	4313	3390	4960	4313	3390
Controls	✓	✓	✓	✓	✓	✓

Notes: This table show the estimation results of equations (4). The dependent variable is the expected social security wealth, its log, or its ratio with income. Standard errors in parentheses and * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Substitutability from SSW to Private Pension Plans

	(1)	(2)	(3)	(4)
Exp. SSW (in €1000)	-0.42**	-0.59***		
	(0.22)	(0.22)		
Exp. SSW / Y			-6.11*	-12.35***
			(3.61)	(4.70)
Observations	4960	4313	4960	4313
Controls	✓	✓	✓	✓
F-stat 1st stage	12.30	15.24	21.66	18.03
Sample	All	Employees	All	Employees

Notes: This table show the estimation results of equations (3), estimated by means of 2SLS. The dependent variable is a dummy equal to 1 if the respondent holds a private pension plan. Standard errors are in parentheses, and * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

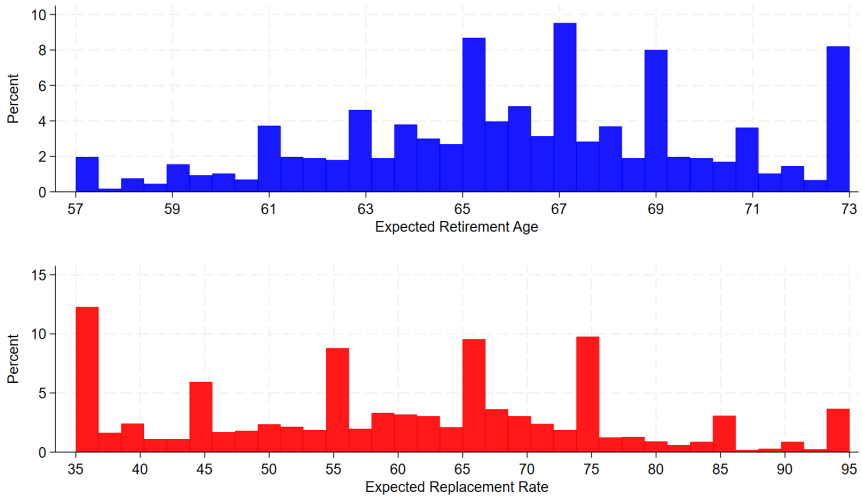
Table 7: Split Sample by Financial Literacy

	(1)	(2)	(3)	(4)
Treat	2.98*	3.57	3.28**	8.49***
	(1.54)	(2.20)	(1.67)	(2.38)
Observations	2975	1985	2609	1704
Sample	All	All	Employees	Employees

Notes: This table shows the estimation results of the equation: $PrivatePension_{it} = \alpha + \beta Treat_i + \varphi \mathbf{X}_i + \lambda_t + u_i$, estimated for low levels of financial literacy (odd numbers) and high levels of financial literacy (even numbers). Standard errors in parentheses and * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

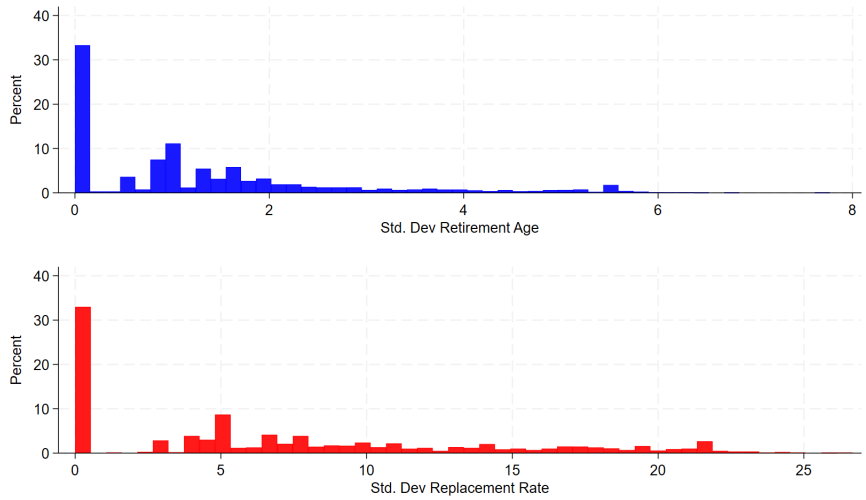
Figures

Figure 1: Expected Retirement Age and Replacement Rate



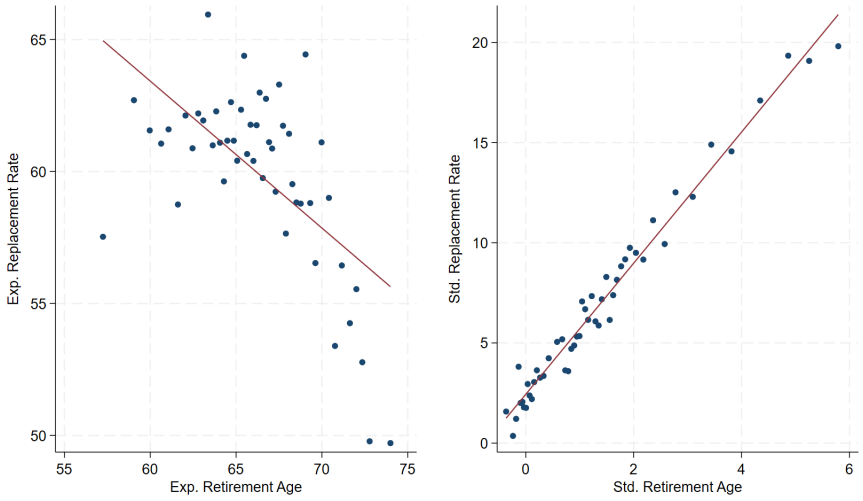
Cross-sectional distribution of the implied means of the subjective distributions of expected retirement age

Figure 2: Standard Deviation of Retirement Age and Replacement Rate



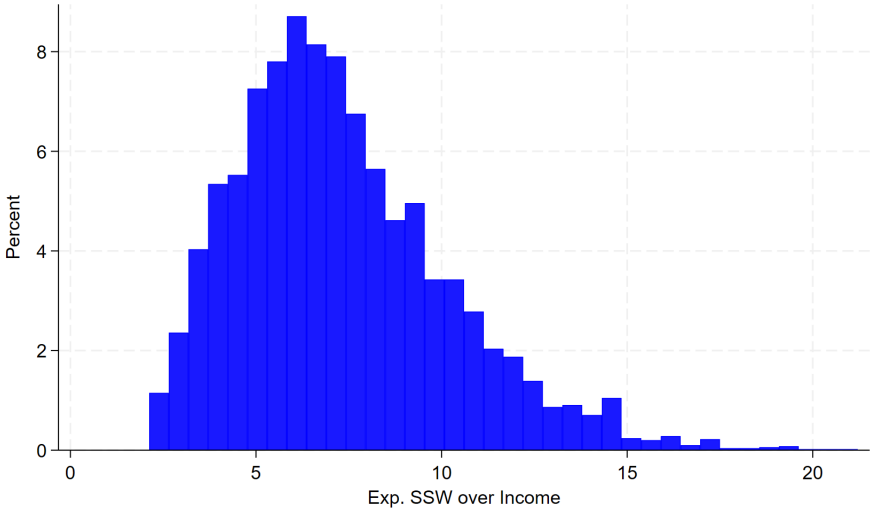
Cross-sectional distribution of the implied standard deviations of the subjective distributions of expected retirement age

Figure 3: Correlation of Expectations



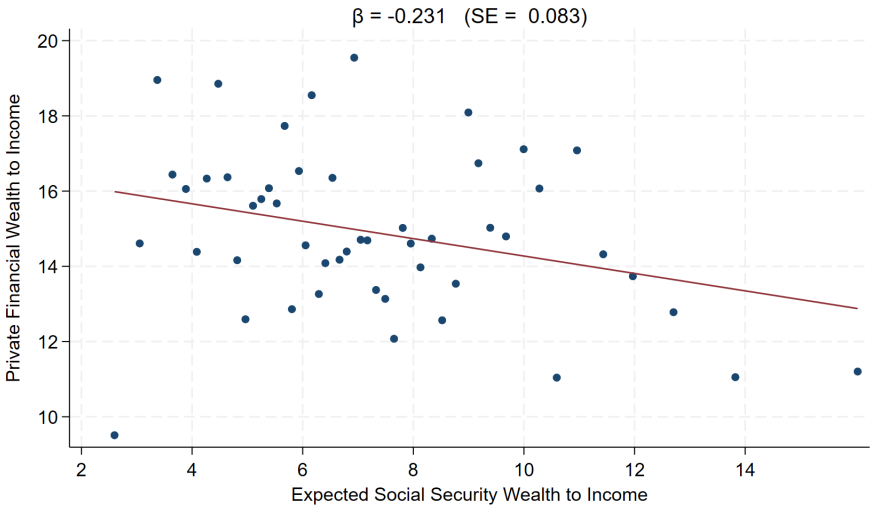
Correlation between the expected retirement age and the expected replacement rate (left panel) and the standard deviations

Figure 4: Distribution of Expected Social Security Wealth



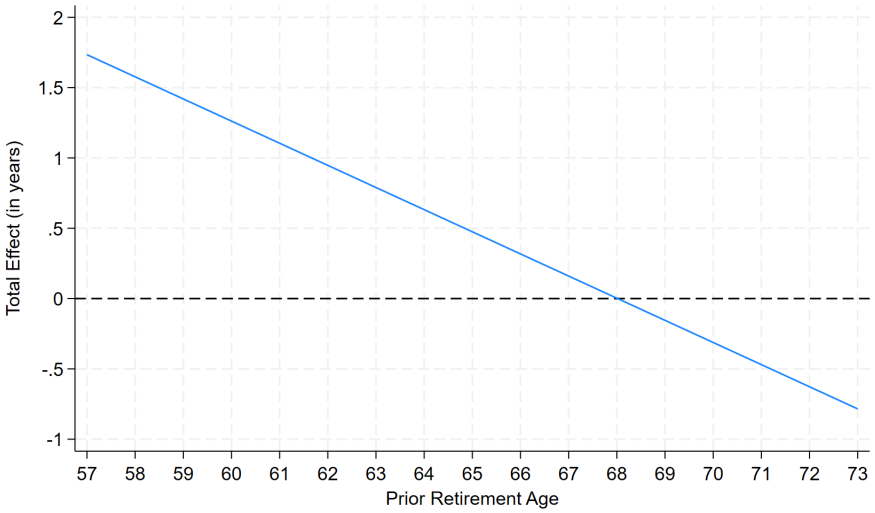
The pooled cross-sectional distribution of Exp. SSW over Income

Figure 5: Correlation between SSW and Private Financial Wealth



This graph shows that there is a negative correlation between the private wealth ratio and private wealth, both scaled down by income. However, this correlation is mild and well below -1.

Figure 6: Treatment Effect as a Function of the Prior



This graph shows the total effect of the treatment as a function of the prior expected retirement age.

Appendix A: The Questionnaire

Section A: Socio-Economic Variables

[label*=A. 0]Gender

1. (a) Male
   (b) Female

2. Age

3. Indicate your level of education
   (a) PhD/Master
   (b) Bachelor's Degree
   (c) University (without degree)
   (d) High School (with certificate)
   (e) High School (without certificate)
   (f) Middle School (with certificate)
   (g) Middle School (without certificate
   (h) Primary School/No Education

4. Indicate the Occupation/Employment Status
   (a) worker or similar position
   (b) employee
   (c) teacher
   (d) managerial/middle management employee
   (e) headmaster, senior official, educational director, university lecturer, magistrate
   (f) freelancer
   (g) entrepreneur
   (h) Self-employed/Craftsman
   (i) Other Independent
   (j) Looking for a first job
   (k) Unemployed
   (l) Housewife

(m) Wealthy

(n) Retired

(o) Student

(p) Doesn't indicate/Reject

5. Can you tell me your marital status?

   (a) Married or in a civil union

   (b) Single

   (c) Separated/Divorced

   (d) Widow/Widower

6. Indicate how many people are in your household, **including yourself**

   (a) 1

   (b) 2

   (c) 3

   (d) 4

   (e) 5

   (f) 6 and more

Section B: Income and Work

[label*=B. 0]**To all:** Considering all of your household's income, what was your total household monthly income in [...], after all taxes?

1. (a) 500-1000€

   (b) 1000-1500€

   (c) 1500-2000€

   (d) 2000-2500€

   (e) 2500-3000€

   (f) 3000-4000€

   (g) 4000-5000€

   (h) 5000-7500€

   (i) 7500-10000€

   (j) 10000-15000€

   (k) over €15000

   (l) Don't know/Don't indicate

2. **To all:** Consider now your earned or retirement income. What was your monthly income in [...], after all taxes?

(a) 500-1000€

(b) 1000-1500€

(c) 1500-2000€

(d) 2000-2500€

(e) 2500-3000€

(f) 3000-4000€

(g) 4000-5000€

(h) 5000-7500€

(i) 7500-10000€

(j) 10000-15000€

(k) over €15000

(l) Don't know/Don't indicate

3. **If employed (a-i in A.4)**: With reference to your current employment situation, indicate how likely you are to be able to keep your job in the next 12 months if you wanted to (*Note: Use a scale of 1 to 100 to answer your answer, with 1 being "unlikely" and 100 being "extremely likely*)

Section C: Household's Wealth

[label*=C. 0]The house you live in is:

1. (a) Owned by my household

(b) For rent

(c) Other (free use or usufruct)

(d) I prefer not to say

2. How much do you think your household's real estate wealth is worth? (owner-occupied dwelling, other dwellings, land)

(a) 0-50000€

(b) 50000-100000€

(c) 100000-200000€

(d) 200000-500000€

(e) 500000-1000000€

(f) Over €1000000

(g) I don't own any real estate

(h) Don't know/Don't indicate

3. How much does your household save? Please think not only about your savings in your checking account but also about any investment products and insurance policies you own (fixed income securities, bond funds, stocks and equity funds, supplementary pensions, life insurance)

   (a) 0-50000€

   (b) 50000-100000€

   (c) 100000-200000€

   (d) 200000-500000€

   (e) 500000-1000000€

   (f) Over €1000000

   (g) I don't have any savings to spare

   (h) Don't know/Don't indicate

4. Can you tell me how much your household's debts are (mortgages, other debts)

   (a) 0-50000€
   (b) 50000-100000€
   (c) 100000-200000€
   (d) 200000-500000€
   (e) 500000-1000000€
   (f) Over €1000000
   (g) Don't have debts
   (h) Don't know/Don't indicate

Section D: Consumption

[label* = D. 0]Considering all your household's consumption (food and non-food consumption, rent expenses, mortgage/loan payments, insurance, utilities, ...), how much did you spend in [...]?

1. (a) 500-1000€
   (b) 1000-1500€
   (c) 1500-2000€
   (d) 2000-2500€
   (e) 2500-3000€
   (f) 3000-4000€
   (g) 4000-5000€
   (h) 5000-7500€
   (i) 7500-10000€
   (j) 10000-15000€
   (k) over €15000

2. In [...], what was your household's average monthly expenditure on food consumption at home and outside the home?

   (a) 500-1000€
   (b) 1000-1500€
   (c) 1500-2000€
   (d) 2000-2500€

(e) 2500-3000€

(f) 3000-4000€

(g) 4000-5000€

(h) 5000-7500€

(i) 7500-10000€

(j) 10000-15000€

(k) over €15000

Section E: Expectations and Intentions (before treatment)

[label* = E. 0]In the next 12 months, you expect that your household's total annual income, net of all taxes you expect to pay and the transfers you expect to receive from the Government (pensions, child benefits, bonuses, etc.), compared to last year...

1. (a) will decrease by more than 8%

   (b) will decrease between 6 and 8%

   (c) will decrease between 4 and 6%

   (d) will decrease between 2 and 4%

   (e) will decrease between 0 and 2%

   (f) will remain constant

   (g) will decrease between 0 and

   (h) will decrease between 2 and

   (i) will decrease between 4 and

   (j) will decrease between 6 and

   (k) will increase by more than 8%

2. In the next 12 months, you expect that your household's overall consumption (consider ALL expenses: food in and out of the home, housing expenses, clothing, transportation, travel, vacations, etc., ...)

   (a) will decrease by more than 8%

   (b) will decrease between 6 and 8%

   (c) will decrease between 4 and 6%

   (d) will decrease between 2 and 4%

   (e) will decrease between 0 and 2%

(f) will remain constant

(g) will decrease between 0 and 2%

(h) will decrease between 2 and 4%

(i) will decrease between 4 and 6%

(j) will decrease between 6 and 8%

(k) will increase by more than 8%

Treatment (only in the 3rd wave)

"According to the latest Annual Report of INPS, the progressive aging of the population, which increases the number of retirees and reduces the number of active workers, makes it increasingly difficult to finance pensions in all European countries, raising financial stability issues"

Section E: Expectations and Intentions (after treatment)

[label* = E. 0, resume]At what age do you plan to retire?

1. (a) Before 58 years of age
   (b) Between 58 and 60 years of age
   (c) Between 60 and 62 years old
   (d) Between 62 and 64 years of age
   (e) Between 64 and 66 years old
   (f) Between 66 and 68 years old
   (g) Between 68 and 70 years old
   (h) Over 72

2. Think about when you will retire and consider only the public pension, i.e., exclude any pension funds and supplementary pensions. What percentage of your earned income will the government pension represent?

   (a) Less than 40% of your last salary before retirement
   (b) Between 40% and 50% of the last salary received before retirement
   (c) Between 50% and 60% of the last salary received before retirement
   (d) Between 60% and 70% of the last salary received before retirement
   (e) Between 70% and 80% of the last salary received before retirement
   (f) Between 80% and 90% of the last salary received before retirement
   (g) Between 90% and 100% of the last salary received before retirement

3. When you are old, do you expect to receive help/support from: (possible multiple answers)

   (a) Sons
   (b) Relatives or friends
   (c) Government
   (d) I will have to provide for myself

[label* = E. 0, resume]In your opinion, what will be the growth of the Italian economy in the next 12 months?

1. (a) will decrease by more than 8%
   (b) will decrease between 6 and 8%

(c) will decrease between 4 and 6%

(d) will decrease between 2 and 4%

(e) will decrease between 0 and 2%

(f) will remain constant

(g) will decrease between 0 and 2%

(h) will decrease between 2 and 4%

(i) will decrease between 4 and 6%

(j) will decrease between 6 and 8%

(k) will increase by more than 8%

2. In your opinion, what will be the inflation rate, i.e., the rate of price growth, in the Italian economy over the next 12 months?

(a) will decrease by more than 8%

(b) will decrease between 6 and 8%

(c) will decrease between 4 and 6%

(d) will decrease between 2 and 4%

(e) will decrease between 0 and 2%

(f) will remain constant

(g) will decrease between 0 and 2%

(h) will decrease between 2 and 4%

(i) will decrease between 4 and 6%

(j) will decrease between 6 and 8%

(k) will increase by more than 8%

3. What do you think the unemployment rate will be in a year?

(a) It will be between 0 and 2%

(b) It will be between 2 and 4%

(c) It will be between 4 and 6%

(d) It will be between 6 and 8%

(e) It will be between 8 and 10%

(f) It will be between 10 and 12%

(g) It will be between 12 and 14%

(h) It will be greater than 14%

4. In a year, at what interest rate do you think you will be able to invest your savings in the financial markets?

   (a) It will be between 0 and 2%

   (b) It will be between 2 and 4%

   (c) It will be between 4 and 6%

   (d) It will be between 6 and 8%

   (e) It will be greater than 8%

5. In a year, what do you think the mortgage interest rate for buying a home will be?

   (a) It will be between 0 and 2%

   (b) It will be between 2 and 4%

   (c) It will be between 4 and 6%

   (d) It will be between 6 and 8%

   (e) It will be greater than 8%

Appendix B: Elicitation of the mean and the standard deviation of the subjective distributions

Take, for example, the question on expected consumption growth:

"In the next 12 months, you expect that your household's overall consumption (consider ALL expenses: food in and out of the home, housing expenses, clothing, transportation, travel, vacations, etc., ...)"

[label=·,nosep] will decrease by more than 8%: X_1 ($p_1 = -9$) will decrease between 6 and

8%: X_2 ($p_2 = -7$) will decrease between 4 and 6%: X_3 ($p_3 = -5$) will decrease between 2 and 4%: X_4 ($p_4 = -3$) will decrease between 0 and 2%: X_5 ($p_5 = -1$) will remain constant: X_6 ($p_6 = 0$) will decrease between 0 and 2%: X_7 ($p_7 = 1$) will decrease between 2 and 4%: X_8 ($p_8 = 3$) will decrease between 4 and 6%: X_9 ($p_9 = 5$) will decrease between 6 and 8%: X_{10} ($p_{10} = 7$) will increase by more than 8%: X_{11} ($p_{11} = 9$)

Total.....100

Thus, $\sum_i X_i = 100$. The expected value is:

$$\mathbb{E}\left[\%\Delta y_i\right] = \sum_i p_i \, X_i$$

Where p_i is the midpoint of each interval and X_i is the probability attached to each bin. On the other hand, the standard deviation is equal to:

$$Std\left[\%\Delta y_i\right] = \sqrt{Var\left[\%\Delta y_i\right]} = \sqrt{\mathbb{E}\left[(\Delta y_i^2)\right] - \left(\mathbb{E}\left[\Delta y_i\right]\right)^2} =$$

$$= \sqrt{\sum_i p_i^2 \, X_i - \left(\sum_i p_i \, X_i\right)^2}$$

For the other questions, compute both moments the same way.